Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 5G nr. 6.5

Gegeven: $f(x) = 2x - x^2$ en $g(x) = x$

1. Snijpunten

$$2x - x^2 = x$$

$$x - x^2 = 0$$

$$x(1 - x) = 0$$

$$x = 0 \text{ en } x = 1$$

2. Bovenste functie bepalen

$$x = \frac{1}{2}$$

$$\begin{cases} f\left(\frac{1}{2}\right) = 2 \cdot \frac{1}{2} - \left(\frac{1}{2}\right)^2 = \frac{3}{4} \\ g\left(\frac{1}{2}\right) = \frac{1}{2} \end{cases}$$

$\Rightarrow f$ boven g

3. Volume

$$V = \pi \int_0^1 \left((2x - x^2)^2 - x^2\right) dx$$

$$= \pi \int_0^1 (4x^2 - 4x^3 + x^4 - x^2) dx$$

$$= \pi \int_0^1 (3x^2 - 4x^3 + x^4) dx$$

$$= \pi \left[\frac{3x^3}{3} - \frac{4x^4}{4} + \frac{x^5}{5}\right]_0^1$$

$$= \pi \left[x^3 - x^4 + \frac{x^5}{5}\right]_0^1$$

$$= \pi \left(\left(1 - 1 + \frac{1}{5}\right) - (0 - 0 + 0)\right)$$

$$= \frac{\pi}{5}$$

x	0		1		2
$f(x)$	0		$+$		0
$f'(x)$	$+$		0		$-$
$f''(x)$	$-$		$-$		$-$
$f(x)$	0	$\nearrow$	$M(1)$	$\searrow$	2
		$\frown$		$\frown$	

x	0	
$g(x)$	0	$+$
$g'(x)$	$+$	
$g(x)$	0	$\nearrow$

5G-6.5-siebe.mees.INF.png

References